Estimator properties

Statistics and Data Analysis

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1 Task

The task consists in estimating the location parameter of the normal, Cauchy and exponential distributions using three estimators for central tendency: mean, trimmed mean and median.

To check the influence of the sample size the whole procedure should be repeated for samples of 10, 100, 1000, and 10000 observations generated from each of the test distributions. Repeat the fitting procedure 100 times for each estimator and each sample size to report the mean and standard deviation of a given estimator. Next, discuss which of them could be

(a) consistent,

(b) unbiased,

(c) effective

by filling Table 1 available in the file `Estimator_properties.ods`. To facilitate interpretation add colours as the cells background conditional on their values. Note that the properties of an estimator such as unbiasedness or consistency are in general derived by means of mathematical proofs. Simulation experiments can only give a partial answer to this question in the case that a given estimator clearly does not meet a given requirement.

2 Tips

- Generation of random samples in R: `rnorm, rcauchy, rexp`.
- For normal distribution with mean $m = 0$ the expectation is 0. For exponential distribution with rate λ it is $1/\lambda$. For Cauchy distribution the expectation does not exist but this distribution is symmetric around 0.
- If you store results in data frame option `options(stringsAsFactors = FALSE)` can be useful.

- Saving results to CSV: `write.table(x = , file = "output.csv", sep = ";", row.names = FALSE, dec = ",")`.
- For exponential distribution with rate λ median equals $\log(2)/\lambda \approx 0.69/\lambda$.
- Look for conditional formatting in the spreadsheet application.

Table 1: Table for reporting results

		Mean of the estimator				Standard deviation of the estimator				Plausible properties		
Distribution	Estimator	$n = 10$	$n = 100$	$n = 1000$	$n = 10000$	$n = 10$	$n = 100$	$n = 1000$	$n = 10000$	Consistent	Unbiased	Effective
normal	mean trimmed mean median											
Cauchy	mean trimmed mean median											
exponential	mean trimmed mean median											